

PrivRobust-SL: a privacy- preserving and adversarially robust split learning framework for IoT intrusion detection

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ABSTRACT

The rapid expansion of Internet of Things (IoT) networks has intensified the demand for machine learning solutions that ensure both privacy and security. Traditional centralised learning models are highly vulnerable to privacy breaches and adversarial manipulations, pointing to the importance of distributed frameworks with integrated defence mechanisms. This work presents PrivRobust-SL, a novel split learning framework that jointly enforces differential privacy and adversarial robustness against privacy leakage and evasion attacks. In PrivRobust-SL, Gaussian differential privacy noise is applied to client-side activations, while the server employs FGSM and PGD-based adversarial perturbations within the joint Denoising Auto encoder (DAE) classifier to enhance feature stability. Experimental evaluations show that PrivRobust-SL maintains a high clean accuracy of 85.2% with only marginal utility loss of 1–2% while improving adversarial accuracy by over 15% at stronger perturbation levels. It achieves 74.22% (FGSM) and 73.10% (PGD) accuracies at $\epsilon = 0.05$, offering robustness on par with AT-only and far exceeding DP-only performance. The higher reconstruction error shows enhanced privacy protection against feature inversion attacks. Overall, the proposed framework demonstrates balanced trade-off among privacy, utility, and robustness, maintaining stable detection accuracy under adversarial attacks, offering a reliable solution for secure IoT environments.

1. Introduction

The Internet of Things (IoT) has grown very rapidly and changed many industries, including healthcare, manufacturing, smart cities, and agriculture. However, this hyper-connectivity also brings huge cyber security vulnerabilities [1]. IoT networks consist of an extensive range of resource-limited devices that frequently lack adequate built-in security measures, rendering them vulnerable to advanced cyber-attacks. Intrusion Detection Systems (IDS) that use machine learning is now essential for detect malicious traffic patterns while keeping IoT infrastructures safe. Deep learning (DL) has performed exceptionally well in various fields, including computer vision, natural language processing, intrusion detection, and anomaly detection [2]. To accomplish conventional DL properly, the data need to collect from one or more sources and trained in one place, especially in the cloud. But this form of deep learning generates major privacy issues and makes it difficult to stop exposing of the raw data to build reliable models [3]. In deep learning, it is essential to preserve a balance between accuracy and privacy by restricting

unauthorized centralised data acquisition [4].

Split learning (SL) divides layers and distributes them to both the client and the central server based on split point selection. The local client performs computation on the first layers before split point, while the server [5] does the remaining part of the layers. Privacy threats are an important concern for split learning systems. In SL, malicious attackers can utilize shared smashed data or intermediate activations to steal private information like raw user data or model parameters, which poses the system's privacy at risk [6]. Addressing Model inversion attacks is extremely important because these allow attackers use intermediate representations or output predictions from the server to reconstruct private input data without having access to it [7]. These attacks illustrate how simple it is to predict sensitive information without actually accessing it, which is a significant issue for IoT and cloud-edge deployments in the real world. To deal with these concerns about split learning, various privacy-preserving methods have been investigated in the previous studies. Homomorphic encryption is one method that keeps data safe and correct and allows people to make

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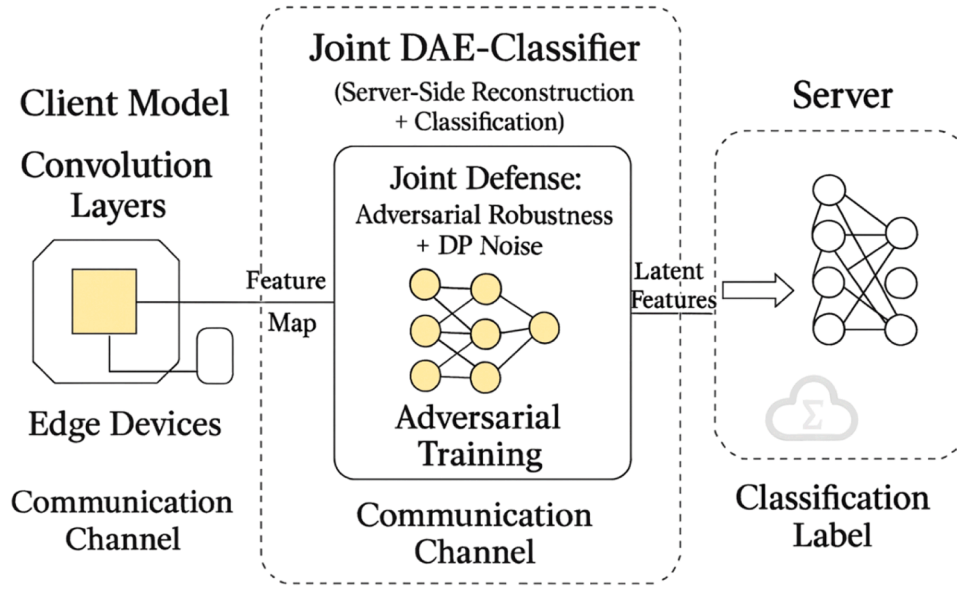


Fig. 1. Architecture of proposed Framework.

calculations on encrypted data without need to decrypt it first [8]. Local differential privacy (LDP) protects smashed data even more by changing it before sending it to the server [9]. This solution is light weight and works well with IoT devices that have limited computational power. But because the noise is added at random, it will always cause a difference in the reconstructed or performance related results, which results trade-off between privacy and utility.

Along with privacy issues, adversarial machine learning has shown IDS well planned changes that induce misclassification but are not noticeable to people [10]. The detection rates can be significantly reduced by the malicious perturbations that are created using methods such as the Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), or Universal Adversarial Perturbations (UAP)[11]. Adversarial training has been thoroughly examined as an effective defence against this problem, as it strengthens models by integrating adversarially modified inputs throughout the training process [12]. Recent studies have demonstrated the importance of robustness in network security applications by incorporating adversarial defences with feature selection [13], generative adversarial networks (GANs) [14], or robust transfer learning methodologies [15]. The presence of both privacy leakage and adversarial evasion makes IoT networks extremely vulnerable and recent IDS frameworks only deal with one of these problems at a time. For example, DP-centric studies protect data exchange but are still open to malicious traffic [16]. On the other hand, defences that focus on robustness make systems stronger but don't stop shared representations from being disclosed [17].

These types of solutions don't work for IoT contexts where both privacy and robustness are very important. Existing intrusion detection systems that use differential privacy are designed to limit feature leakage; however, they frequently suffer significant accuracy deterioration in the presence of acceptable noise levels and remain susceptible to adversarial manipulation. On the other hand, adversarial training approaches make systems more resistant to evasion attacks, but they fail to address privacy leaks at the client-server interface in the context of split learning. Because of this, previous IDS frameworks based on split learning fail to balance privacy, robustness, and detection utility in a consistent way.

To address this gap, this work proposes a unified architecture that jointly incorporates adversarial training and differential privacy noise injection on the client side of smashed data. This is done through a shared denoising auto encoder (DAE) classifier on the server. The DAE not only reduces the distortion caused by DP perturbations but also

generates a reconstruction-error signal that improves adversarial detection. We validate our methodology using the CICIDS2018 dataset, a widely used benchmark for assessing IDS in realistic attack scenarios. The main objective of this work is to develop split learning based intrusion detection system that protects privacy and resistance against adversarial attacks.

The main research objectives of this study are:

- To establish the trade-off between privacy, accuracy, and robustness in split learning
- To examine the efficacy of DP-protected intrusion detection models against adversarial-attacks.
- To develop a unified framework that combines differential privacy, adversarial training

With denoising auto-encoder to improve robustness while preserving privacy.

Fig. 1 shows the full architecture of the proposed system, which combines differential privacy and adversarial robustness utilising a joint DAE classifier on the server side.

The major contributions of the work are:

- Framework design:** We introduce PrivRobust-SL, an IoT-focused split learning IDS that combines adversarial training with differential privacy noise injection.
- Methodological innovation:** We develop a combined denoising auto encoder (DAE) Classifier, in which reconstruction error acts as both a regularizer and a detector of Adversarial examples.
- Thorough assessment:** We perform extensive evaluation on the CICIDS2018 dataset, thoroughly investigating the relationship between privacy budgets, adversarial attack budgets.

2. Related work

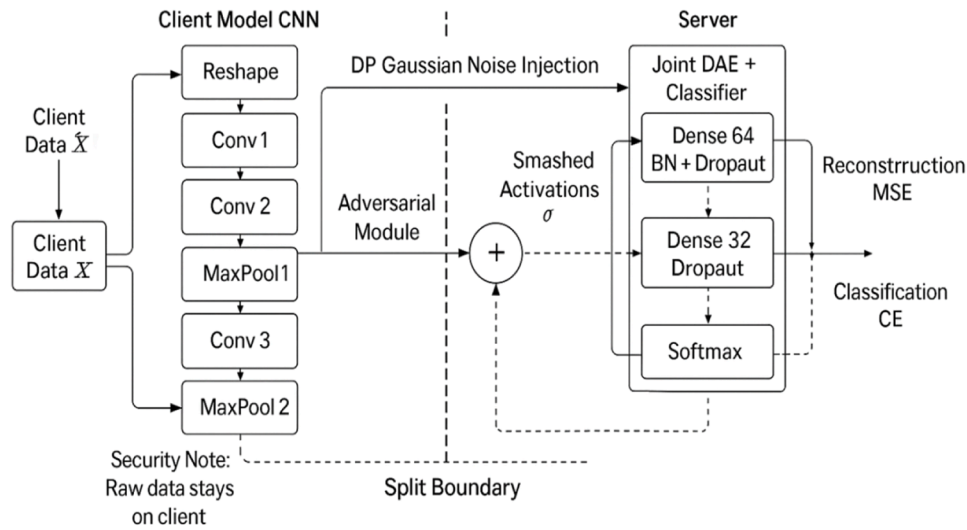
2.1. Differential Privacy in Split and Federated Learning

Split learning (SL) has emerged as a distributed paradigm for training deep models without direct exposure to raw data, making it a natural fit for privacy-sensitive IoT applications. Nevertheless, the smashed activations that are transmitted between the client and server are still susceptible to model inversion attacks and reconstruction. Several works have investigated differential privacy (DP) as a defence in order to

Table 1

Comparative analysis of state of art frameworks with quantitative results.

Research Paper	Category	Privacy Mechanism	Robustness Mechanism	Reported Quantitative Results	Key Limitation
Pham et al., [18] IEEE TETCI 2025	DP-only IDS	Differential privacy on intermediate data	Not addressed	Accuracy drops from 99% to 66-73% for DP enabled clients($\epsilon = $, indicating 25-30% loss in utility)	Adversarial Robustness not evaluated; IDS attacks not considered
Mahawaga et al., [21]	DP-only IDS	Local differential privacy (LATENT)	Not addressed	Maintains 91-96% accuracy only at relatively high ϵ	No adversarial threat analysis
Pham et al.,[20] (2023)	SL with privacy Enhancement	Binartization + Optional DP	Feature hijacking defense	Reports ≤ 2 -3% accuracy loss compared to full-precision SL on benchmark tasks	IDS-related adversarial robustness not studied
Alhussien et al.,[25]	AT-only IDS	Not Considered	Adversarial Training (FGSM/PGD)	Adversarial training reduces attack based loss by 20-30% compared to Normal IDS	Privacy leakage not addressed
Yao et al.,[24]	AT-only IDS	Not Considered	Traffic adversarial modeling+ defenses	Adversarial traffic achieves 99% accuracy	Privacy and split learning not considered
PrivRobust-SL (Proposed Work)	DP+AT only	Client-side DP +DAE modelling	Joint Adversarial training	Maintains 85% clean accuracy under DP and improves robustness against FGSM/PGD attacks	Analyses privacy-robustness tradeoff

PrivRobust-SL: Split Learning with DP + Adversarial Training + Joint DAE Classifier**Fig. 2.** Layered architecture of proposed PrivRobust-SL frame work.**Table 2**

Mapping between gaussian noise scale and privacy budget.

Noise scale(σ)	Privacy budget (ϵ)
0.05	96.89
0.10	48.44
0.15	32.29
0.20	24.22
0.25	19.37
0.30	16.14
0.35	13.84
0.40	12.11
0.45	10.76
0.50	9.68
0.55	8.80
0.60	8.07
0.65	7.45

handle model inversion attacks.

Pham et al [18]. performed an experiment to find a balance between privacy and accuracy in split learning using differential privacy. They suggested ways to improve performance of clients cares about privacy and lower the cost of communication. It concludes that using DP noise at lower levels of the network gives better trade-offs and proposes ways to make data smaller. Extensive trials show that the technique works well

with different privacy needs by showing better accuracy and privacy protection across a range of datasets.

In [19], Alhindi et al. presented a novel split learning methodology that combines adversarial training with channel pruning in the model training process to attain a balance between privacy and utility. Experiments on the FairFace and CelebA datasets show that the method lowers the similarity between the original and reconstruction data by more than 50% without significantly lowering the accuracy of the task. This solution provides an improved balance between privacy and usefulness for devices that don't have a lot of resources than the best defences available, such as PAN, DISCO, and noise-based methods. The optimal balance of split point selection and pruning ratios is examined; illustrating that privacy is enhanced while performance is maintained by deeper divides and appropriate pruning ratios.

Ngoc et al [20]. introduced binarised split learning (B-SL), a method that reduces compute and memory costs and enhances data privacy by binarising client-side neural layers. Research indicates that binarization diminishes the capacity to reconstruct data, while differential privacy measures and leakage restriction training enhance privacy. B-SL models provide accuracy that is nearly identical to full precision, with minimal privacy compromises. While local binarised layers prevent leakage, their number should be controlled for performance and privacy. B-SL resists feature-space hijacking and other advanced privacy assaults, enabling

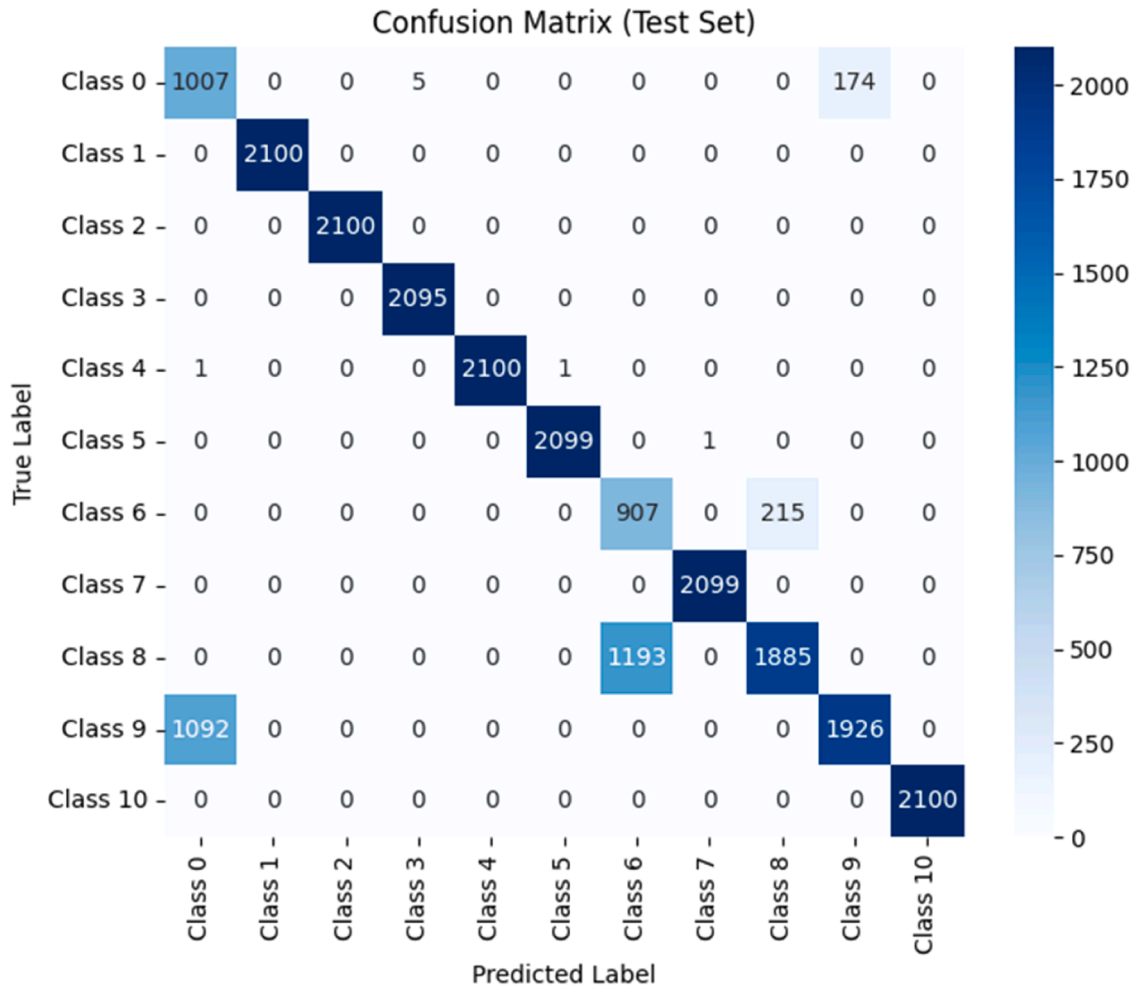


Fig. 3. Confusion matrix for default split learning model.

secure distributed learning applications.

In the work [21], pathum et al. presented LATENT, a local differential privacy (LDP) technique for convolutional neural networks in deep learning models. LATENT's model has three modules: convolutional, randomization, and fully linked layers. For privacy, the client handles randomization. LATENT's architecture reduces data owners' computational workloads, making it suitable for IoT and cloud contexts. It reduces privacy leakage before server-side training by separating the CNN structure and aggressive local randomization. Attacks like membership inference and model inversion risk machine learning privacy, yet the technology is resistant. The paper finds that LATENT improves distributed learning privacy and utility while adapting to future deployments and architectures.

Differential privacy (DP) has been used to keep sensitive model outputs safe in privacy-preserving machine learning inference [22]. This is especially important in clinical settings where leaking inference could put patient privacy at risk. Satellite communication frameworks have integrated DP with SL to enable secure multi-client collaborations with communication limitations, thereby demonstrating the efficacy of split learning. These studies indicate that DP works well to avoid breaches yet it has an accuracy-privacy trade-off that is complicated to fix in IoT intrusion detection, where high fidelity is very important.

2.2. Adversarial Threats and Defenses in Network IDS

In IoT Networks, Malicious traffic detection systems that are developed using machine learning are becoming more vulnerable to

adversarial manipulations, resulting in large-scale misclassifications in Network traffic features. The exhaustive survey of adversarial challenges in NIDSs has gained significance with gradient-based methods, such as FGSM and PGD, can degrade detection rates.

Holla et al [23]. proposed a dual-layered defence that combines adversarial training (designed assault samples) and SHAP-based robust feature selection to enhance accuracy, interpretability, and resilience. The proposed feature selection addresses the vulnerability of protocol and statistical traffic aspects to attack manipulation. In the CSE-CICIDS2018 investigations, the combined defence mechanism reduces attack success rates by 40% and enhances IDS accuracy to 88% for gradient-based and 84% for black-box (Square Attack) adversaries. Real-time deployment of inference is enabled by the defence, which increases the difficulty of training by 25%. The study recommends federated learning, hybrid, and detection-based research, and the approach is generalisable.

Yao et al [24]. introduced Packet-level Adversarial Traffic Generation (PATG) to generate realistic adversarial IoT traffic that circumvents deep learning-based NIDS while maintaining malicious capability. It generates protocol-compliant traffic perturbations by employing reversible traffic vectorisation and PS-WGAN, a Wasserstein GAN-based model. PATG exhibits evasion rates exceeding 50% on nine DL-based NIDS across multiple IIoT datasets, all with modest execution costs. This approach is effective in gray-box and closed-box attacks, requiring minimal prior knowledge. Adversarial training and feature selection can prevent attacks while maintaining a tradeoff between computational efficiency and robustness. The next phases involve strengthen defences

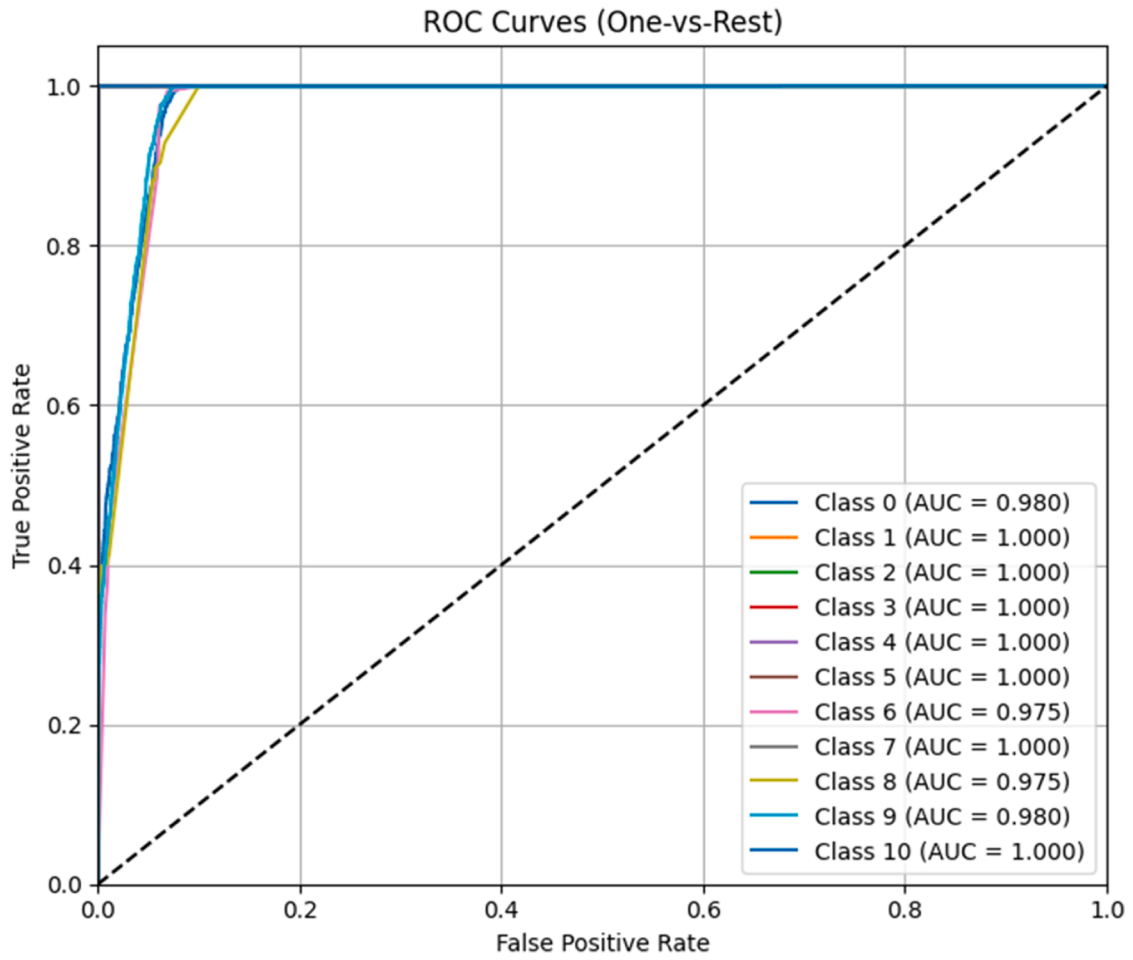


Fig. 4. ROC-AUC Curves for the default split learning model.

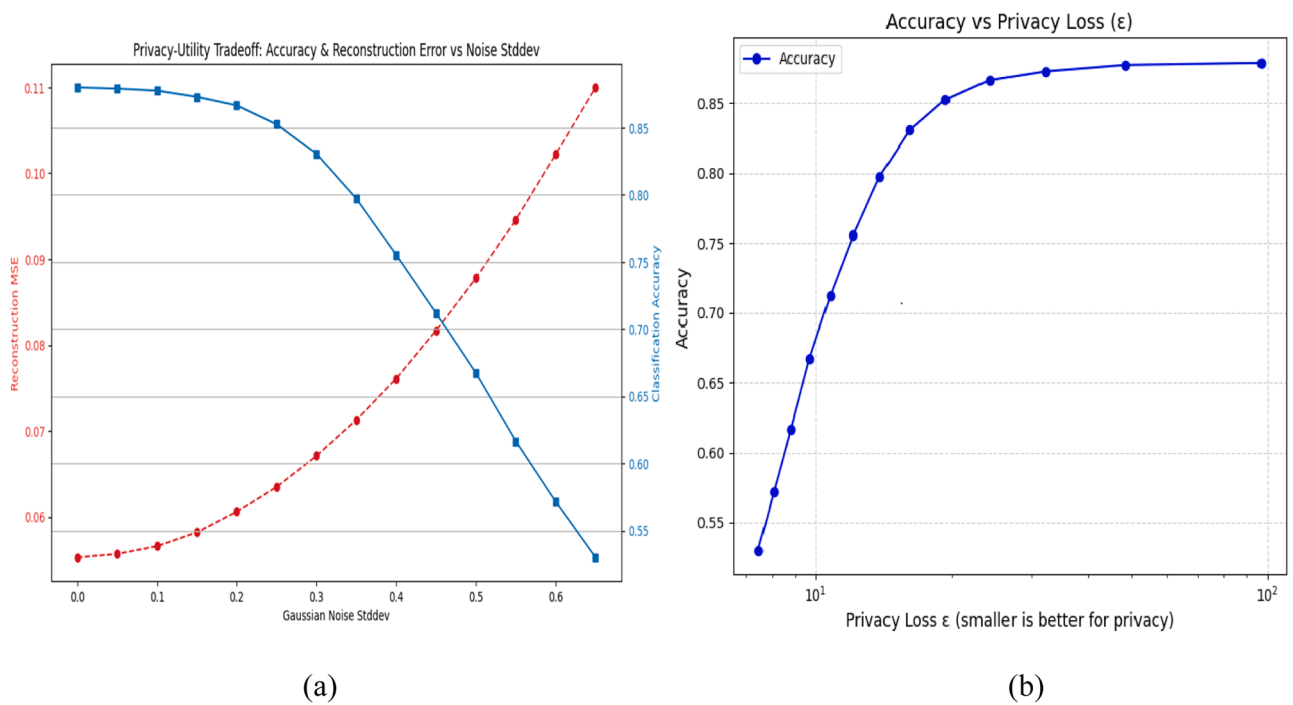
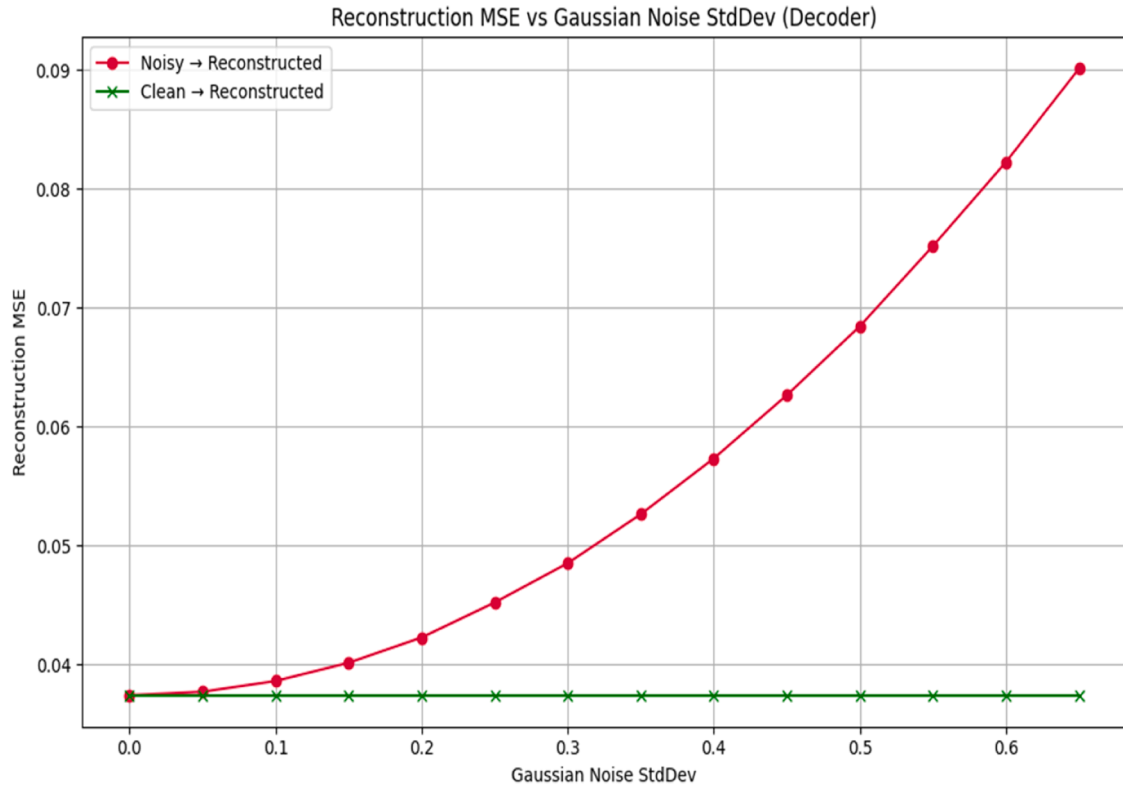
Fig. 5. Privacy vs utility tradeoff for various noise levels and variation of accuracy with ϵ .

Table 3Accuracies and reconstruction error across σ .

Noise scale(σ)	0.0	0.05	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5	0.55	0.6	0.65
Accuracy (%)	88.1	88.1	87.8	87.2	86.6	85.3	82.7	79.6	75.8	71.1	66.8	61.6	57.6	53.3
Reconstruction error (%MSE)	5.53	5.57	5.6	5.8	6.0	6.3	6.7	7.1	7.6	8.1	8.7	9.4	10.2	11.0

**Fig. 6.** Reconstruction error For clean vs noisy activations.

against adversary traffic and adapting to dynamic contexts.

The significance of domain-aware constraints for adversarial traffic has been underscored in other studies. Alhussien et al [25]. observed that adversarial training provided partial resilience, but transferability remained a challenge, and therefore constrained perturbations to maintain valid traffic semantics. Beyond evasion, adversarial manipulations and membership inference assaults are increasingly being examined in conjunction. Hierarchical Differential Privacy was proposed by Liu et al [26]. to decrease the success rates of both adversarial attacks and membership inference in traffic classifiers, resulting in a reduction of over 60% in attack success. Youssef et al [27]. observed adversary training arises as an iterative defence strategy in IoT-enabled smart grids, where dynamic load-altering attacks further illustrate how adversarial perturbations can destabilize physical systems.

Differential privacy for split learning and adversarial robustness in IDS has been addressed separately, but few have attempted to incorporate them into a single IoT intrusion detection system. DP-based SL studies mostly analyze privacy–utility trade-offs without adversarial evasion. Adversarial defence research boosts robustness but ignores data smashed privacy leaks. IoT environments must protect sensitive communications against inference attacks and evasion tactics while addressing privacy and robustness.

Table 1 shows the comparative analysis of the works that reported in literature along with quantitative information. Current IDS frameworks may be generally classified into DP-only and AT-only methodologies. DP-only methods consider privacy first, but they often suffer from accuracy degradation and are vulnerable to adversarial attacks. AT-only

methods improve robustness, but they don't stop feature leakage, especially in distributed or split learning situations. These findings inspire the proposed system, which integrates differential privacy and adversarial training in a split learning-based IDS.

3. Methodology

PrivRobust-SL is an IDS, which is specifically designed for IoT networks and utilises split-learning. In layered architecture shown in Fig. 2, server in the cloud performs adversarially-robust classification with a joint DAE-classifier that simultaneously reconstructs the original features and classifies network traffic, while a lightweight client on the edge transforms raw CICIDS2018 [28]. features into a compact representation. Adversarial training is conducted at the server, while differential privacy (DP) is enforced at the client before transmission. CICIDS 2018 per-flow features are provided to the client side as input with 37. On the client side of the model, features are transformed into a 1D sequence and passed through a compact CNN stack, which consists of two convolution layers with 64 filters each, followed by a max-pooling layer. Subsequently, a third convolution layer with 128 filters is implemented, followed by an additional pooling layer. The denoising autoencoder (DAE) is built as a symmetric, completely connected structure that works on the split-layer representation on server side. There are two dense layers in the encoder: a hidden layer with 64 units and ReLU activation, and a latent layer with 32 units and ReLU activation. After the hidden layer, batch normalization and dropout (rate = 0.2) are used to make training more stable and generalize better.

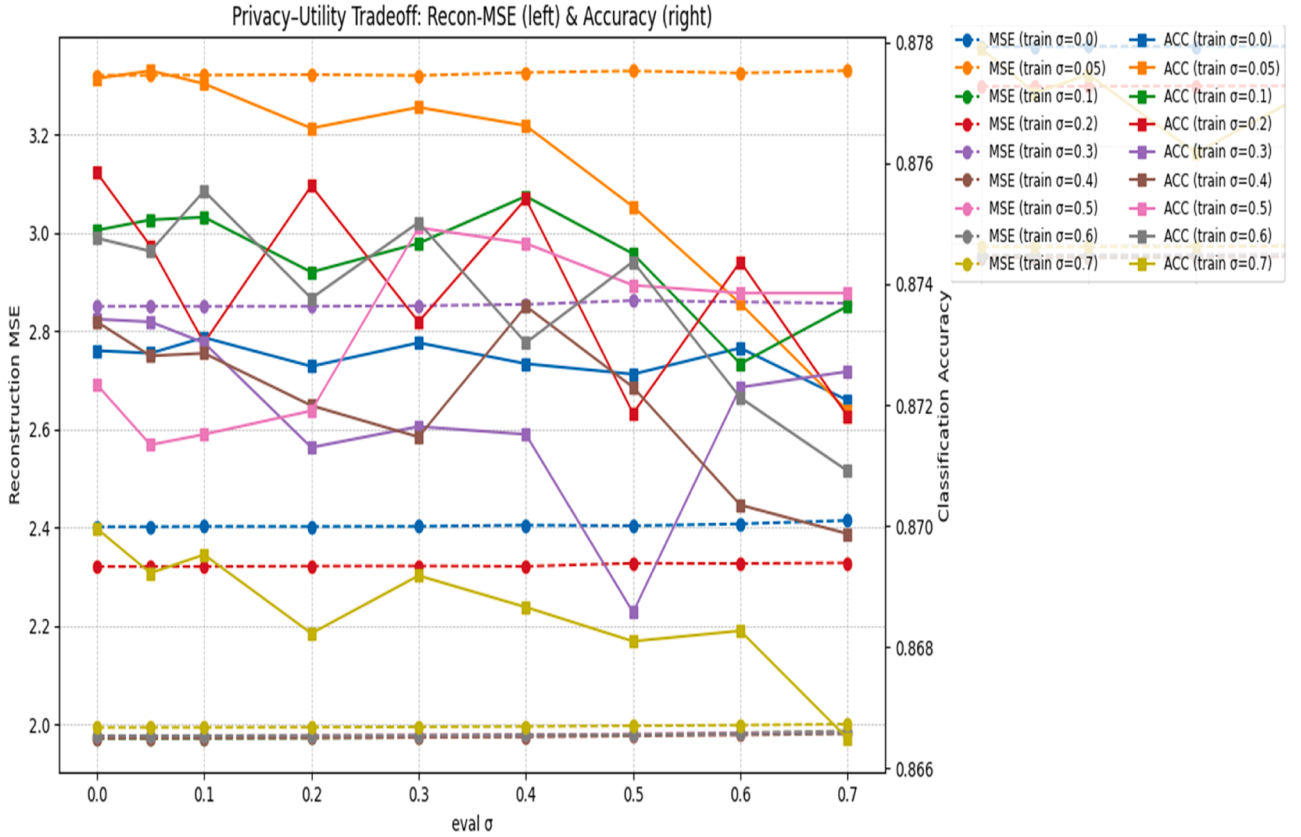
Fig. 7. MSE vs accuracy at fixed σ_{train} evaluated across all σ_{eval} .

Table 4
Generalization across noise mismatch.

Train σ	Eval σ	Accuracy (%)	Recon. MSE
0.0	0.0	87.29	2.402
0.0	0.3	87.30	2.402
0.0	0.5	87.26	2.403
0.2	0.0	87.31	2.328
0.2	0.3	87.56	2.329
0.2	0.5	87.48	2.330
0.4	0.0	87.32	1.980
0.4	0.3	87.35	1.982
0.4	0.5	87.22	1.983

The Adam optimizer with a learning rate of 0.001 is used to train all of the models. All experiments employ a batch size of 256, and training may continue for up to 50 epochs. To prevent over-fitting, early stopping with a patience of 5 epochs is used based on validation loss. We use categorical cross-entropy as the loss function for classification.

The output of client block is the smashed activation tensor z , which is flattened $\tilde{z}$ and prepared for transmission to the server by applying differential privacy (DP) noise. The remaining layers of the network incorporate adversarial augmentation together with a joint denoising autoencoder (DAE) classifier on the server side. At this stage, the server is responsible for both classifying the incoming traffic flows and reconstructing the original feature representations from the perturbed activations, thereby ensuring robust detection while preserving data fidelity.

As both client and server sides contain limited computational parameters, adversarial training and differential privacy techniques impose significant computational overhead primarily during offline training, particularly because of repetitive PGD perturbations and noise injection. These operations are not necessary during inference;

therefore, the latency at deployment time is the same as it would be with regular split learning inference. This separation between training and inference costs makes the suggested method good for IoT applications with limited resources while still offering better privacy and security.

3.1. Differential privacy integration

The smashed activations are added with Gaussian Noise prior to their transmission to server using Differential privacy (DP) technique. This procedure assures that adversaries are unable to directly reconstruct or exploit sensitive information contained within the features. The primary justification is that the IoT device retains raw traffic data indefinitely, while only DP-perturbed intermediate representations are transmitted to the server. This approach mitigates both privacy risks and communication expenses.

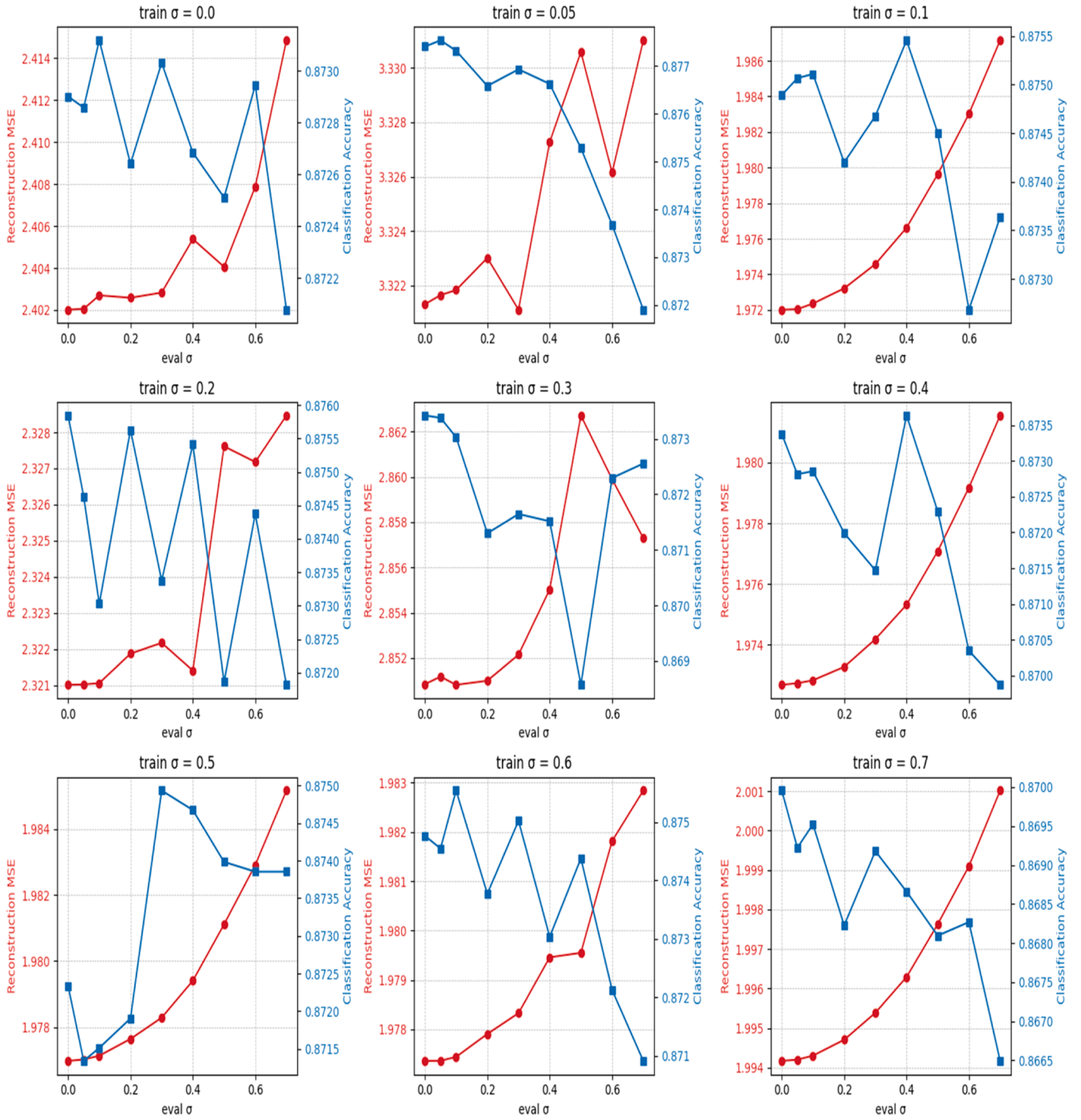
Element wise Gaussian mechanism on $\tilde{z}$ is shown in [equation 1](#):

$$\hat{z} = \tilde{z} + \mathcal{N}(0, \sigma^2) \quad (1)$$

Where σ denotes the noise standard deviation, which determines the privacy budget. The injected noise ensures that any single data sample's contribution is indistinguishable, thereby limiting the feasibility of reconstruction and re-identification attacks. The corresponding privacy budget [29]. calculation for σ is obtained as in [equation 2](#):

$$\epsilon = \frac{S \cdot \sqrt{2 \ln \left(\frac{1.25}{\delta} \right)}}{\sigma} \quad (2)$$

Where 'S' represents the sensitivity of the mechanism and δ is the failure probability. In our work the values of 'S' and ' δ ' are fixed as 1.0 and 10^{-5} respectively. As a result, ϵ decreases inversely with σ , signifying that larger noise scales enforce stronger privacy guarantees. To explore the privacy-utility trade-off, multiple noise scales are applied ($\sigma \in \{0.05,$

Privacy-Utility Tradeoff per Train σ Fig. 8. Privacy vs utility at the same noise scale ($\sigma_{train} = \sigma_{eval}$).**Table 5**
Matched training-evaluation noise.

σ (Train = Eval)	Accuracy (%)	Recon. MSE
0.0	87.29	2.402
0.05	87.70	3.326
0.1	87.49	1.975
0.2	87.56	2.329
0.3	87.26	2.858
0.4	87.32	1.980
0.5	87.48	1.978
0.6	87.36	1.982
0.7	86.65	1.994

0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, and 0.65)) and accuracies are plotted for corresponding privacy budget (ϵ).

Lower values preserve higher model accuracy but provide weaker protection, while higher values offer stronger privacy guarantees with some degradation in utility. This systematic variation enables quantifiable analysis of how privacy levels affect detection performance. The Table 2 shows the mapping between Gaussian noise scale (σ) and privacy budget (ϵ).

3.2. Adversarial Attacks Integration

Adversarial samples were initially identified in computer vision and

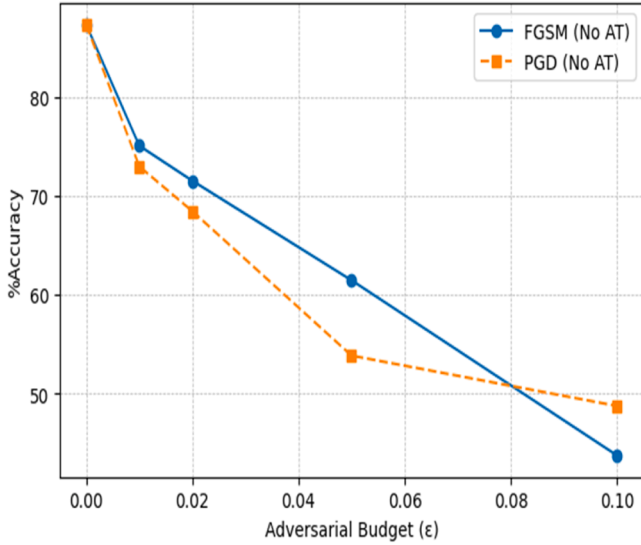


Fig. 9. Accuracy vs. adversarial budget (ϵ) without adversarial training.

Table 6

Accuracy of Joint DAE classifier under FGSM and PGD attack without adversarial training.

Attack	Clean($\epsilon=0$)	$\epsilon=0.010$	$\epsilon=0.020$	$\epsilon=0.050$	$\epsilon=0.10$
FGSM	87.30	75.04	71.50	61.43	43.72
PGD	87.30	72.95	68.42	53.81	48.72

Table 7

Accuracy of Joint DAE classifier after adversarial training.

Attack	Clean($\epsilon=0$)	$\epsilon=0.010$	$\epsilon=0.020$	$\epsilon=0.050$	$\epsilon=0.10$
FGSM(AT)	86.51	82.47	81.23	75.56	66.39
PGD(AT)	86.43	81.72	79.37	74.23	65.60

are now recognized in a variety of domains, such as intrusion detection, malware detection, natural language processing, and speech recognition [30]. Adversarial samples are generated by intentionally introducing minor, deliberate perturbations to legitimate inputs, resulting in an incorrect output being produced by a trained model. Their existence is a significant concern for security-critical applications, such as ML-based intrusion detection systems, as it exposes the intrinsic vulnerabilities of machine-learning models.

The Fast Gradient Sign Method (FGSM) is a simple and efficient one-step attack that uses the gradient of the loss with respect to the input to craft a perturbation [31]. The adversarial sample is generated by equation 3:

$$x^* = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)) \quad (3)$$

Where x is the clean input, ϵ is the perturbation budget, J is the loss, and θ are the model parameters. FGSM is computationally efficient and advantageous for large-scale testing nevertheless; it is generally inferior to iterative methods due to its reliance on a single update.

Projected Gradient Descent (PGD) is an iterative variant of FGSM that takes small gradient steps repeatedly and projects the result back into an allowed neighborhood around the original input [32]. The PGD upgrade is as follows:

Start with a small random perturbation δ around clean input:

$$x^{(0)} = x + \delta,$$

Where δ is sampled randomly within the allowed perturbation region, then iterate T times using a small step α , and after each step project

back into the allowed neighborhood $\beta_\epsilon(x)$. The untargeted PGD update is commonly written as shown in equation 4:

$$x^{t+1} = \Pi_{\beta_\epsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, x^t, y))) \quad (4)$$

PGD usually produces stronger adversarial examples than FGSM and is commonly used for rigorous robustness evaluation. In the activation space received at the server, we explicitly apply both FGSM and PGD in our split-learning setup. Adversarial Training: A curriculum-based feature-space adversarial training technique is used to make the system more resilient to attacks. To keep the accuracy clean, the model is only exposed to minor perturbation magnitudes during training. Adversarial examples based on FGSM are presented with $\epsilon = 0.005$ and $\epsilon = 0.01$. Then, lightweight PGD-based adversarial training is performed with $\epsilon = 0.02$, a constrained iteration budget of $T = 3$ steps, and a step size of $\alpha = \frac{\epsilon}{T}$ (0.067), to ensure cumulative perturbations remains bounded while limiting computational overhead. In each mini-batch, 30% of the samples are substituted with adversarial examples, whereas the other samples remain clean. Adversarial Evaluation: The trained model is evaluated against more severe adversarial perturbations during assessment to check working in the worst-case scenario. We use $\epsilon \in \{0.01, 0.02, 0.05, \text{ and } 0.10\}$ for FGSM and PGD assaults. These higher ϵ values are exclusively utilised to measure robustness loss during inference, not during training.

Algorithm: Adversarial Training in PrivRobust-SL.

Inputs: dataset $D = \{(x, y)\}$, clip norm C , noise scale σ , attack budget ϵ , PGD step size α , PGD steps T , trade-off λ . Outputs: trained parameters θ (client), ϕ (classifier), ψ (decoder)

Initialize parameters θ, ϕ, ψ

For each epoch, iterate over mini-batches of training data, sample mini batch $B = \{(x_i, y_i)\}$ from D

Client forward: compute smashed activations at split point (z_i)

Apply DP at client: Add Gaussian Noise to activations and send to server ($\tilde{z}_i$)

Adversarial augmentation (server): from each received activation, create adversarial variants using FGSM and PGD (respecting feature validity bounds)

Joint DAE-classifier forward (server):

Encode $\tilde{z} \rightarrow h$; decode $h \rightarrow \hat{x}$; classify $h \rightarrow \hat{y}$

Compute losses: combine classification loss (for labels) and reconstruction loss (for decoder) using the chosen λ

Back propagate (server): Update θ, ϕ, ψ with respect to total loss

Back propagate (client): gradients flow back through the cut layer; client-side layers are updated normally

Track performance: record clean vs adversarial accuracy, reconstruction error, and privacy budget determined by the activation noise

The split architecture in IoT settings reduces computation on constrained devices by executing only a lightweight client-side sub-network locally. Adversarial training guarantees resilience against crafted evasion assaults, while DP guarantees the prevention of sensitive traffic feature leakage. When combined, these results in reliable IDS that is appropriate for cloud-edge IoT ecosystems.

4. Results & discussion

The proposed PrivRobust-SL framework is assessed on the CICIDS2018 dataset, which comprises eight main attack categories (Brute Force, DoS, Botnet, DDoS, Web attacks, Infiltration, Port Scan, and Heart bleed) in addition to normal traffic. 37 flow-based features are chosen as model inputs following preprocessing. The dataset is partitioned into three categories: training (70%), validation (10%), and test (20%). One-hot encoding is implemented for labels in nine classes. The proposed system is evaluated across multiple dimensions to provide a holistic understanding of its effectiveness. Robustness is evaluated by contrasting clean accuracy with adversarial accuracy in FGSM and PGD attack scenarios, while detection performance is evaluated in terms of classification accuracy. Reconstruction error analysis is conducted in conjunction with reported (ϵ, σ) - differential privacy guarantees to quantify privacy.

Finally, the system's success in safeguarding IoT environments is

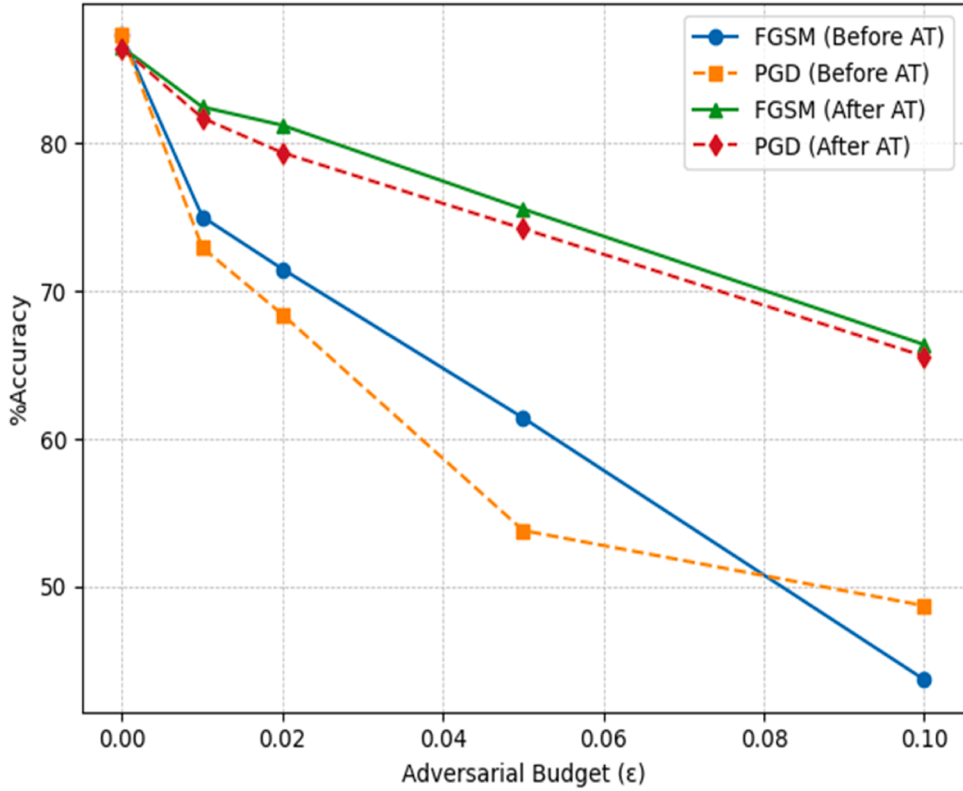


Fig. 10. Accuracy vs. adversarial budget before and after adversarial training.

underscored by the examination of trade-offs between privacy, utility, and robustness in the context of varying noise levels (σ) and adversarial budgets (ϵ).

For the basic Split learning model, the confusion matrix in Fig. 3 shows how effectively each class is identified and misclassification between attack categories that are extremely similar. Also, ROC-AUC analysis has shown in Fig. 4 offers an evaluation of discriminative ability that is not dependent on a threshold. While the overall accuracy reflects performance at a fixed argmax decision threshold, the ROC-AUC curves in Fig. 4 capture probability ranking across all thresholds; consequently, high AUC values indicate strong class separability even when some class overlap reduces overall accuracy.

4.1. Privacy-Utility Trade-off with DP Noise

We initially assess the default split-learning (SL) pipeline in the absence of any server-side Joint DAE. Initially, server consists of two convolutional layers with 128 and 64 filters, followed by a dense layer with 256 filters. The client smashed activations are subjected to Gaussian DP noise with a scale of σ during training and evaluation. The privacy-utility curve (Accuracy vs. σ) is depicted in Fig. 5, while Table 3 provides clean-set accuracy across σ and ϵ . Accuracy was also evaluated against the privacy budget (ϵ), where smaller ϵ values correspond to stronger privacy guarantees. The results show that accuracy decreases as ϵ becomes tighter, while larger ϵ values reduces privacy constraints and preserve higher accuracy. In addition, we calculate the reconstruction error (MSE) between pure activations and their DP-perturbed counterparts to quantify information loss is presented in Table 3. The reconstruction error distributions before and after DP noise are compared in Fig. 6. Accuracy experiences a gradual decline as σ increases, with a more severe decline beyond moderate noise. This is consistent with expectations. The obstruction of client features is confirmed by the fact that the reconstruction error increases monotonically with σ .

The expected trade-off between privacy and utility is shown in the

results. As the Gaussian noise scale (σ) increases, the classification accuracy progressively decreases as a result of the more severe perturbations that are applied to the client activations. The model maintains high accuracy while providing meaningful differential privacy at moderate noise levels ($\sigma \approx 0.2-0.3$), which is when the optimal balance is achieved. The model's utility is diminished when excessive noise is applied, as seen by the significant decrease in accuracy at higher noise levels ($\sigma > 0.5$).

In order to conduct a more thorough examination of the privacy-utility trade-off, we employ a lightweight decoder with two hidden layers of 128 neurons (ReLU) and an output layer that maps to the original activation dimension to analyse reconstruction error. The reconstruction error for pure activations is nearly constant across all σ values, suggesting a faithful recovery. Conversely, smashed activations exhibit a consistent increase in error as σ increases, which is indicative of the increase of learned representation distortion. At moderate noise levels, the increase is incremental and strikes a balance between privacy and utility. However, at higher scales ($\sigma > 0.5$), the error increases rapidly, which not only complicates inversion but also diminishes feature extraction and classification accuracy.

4.2. Differential privacy with Joint DAE classifier

In this section, we introduce the Joint Denoising Autoencoder (DAE) classifier at the server to augment the baseline split learning model. The DAE branch is designed to reconstruct perturbed activations, while the classifier branch simultaneously performs predictions. We trained the Joint DAE classifier with different fixed noise levels (σ_{train}) to study the impact of differential privacy noise under more realistic deployment settings. The Performance of proposed classifier is evaluated across range of evaluation noise scales (σ_{eval}). This setup reveals both the tolerance of the model to mismatched noise distributions and the stability of the privacy-utility trade-off.

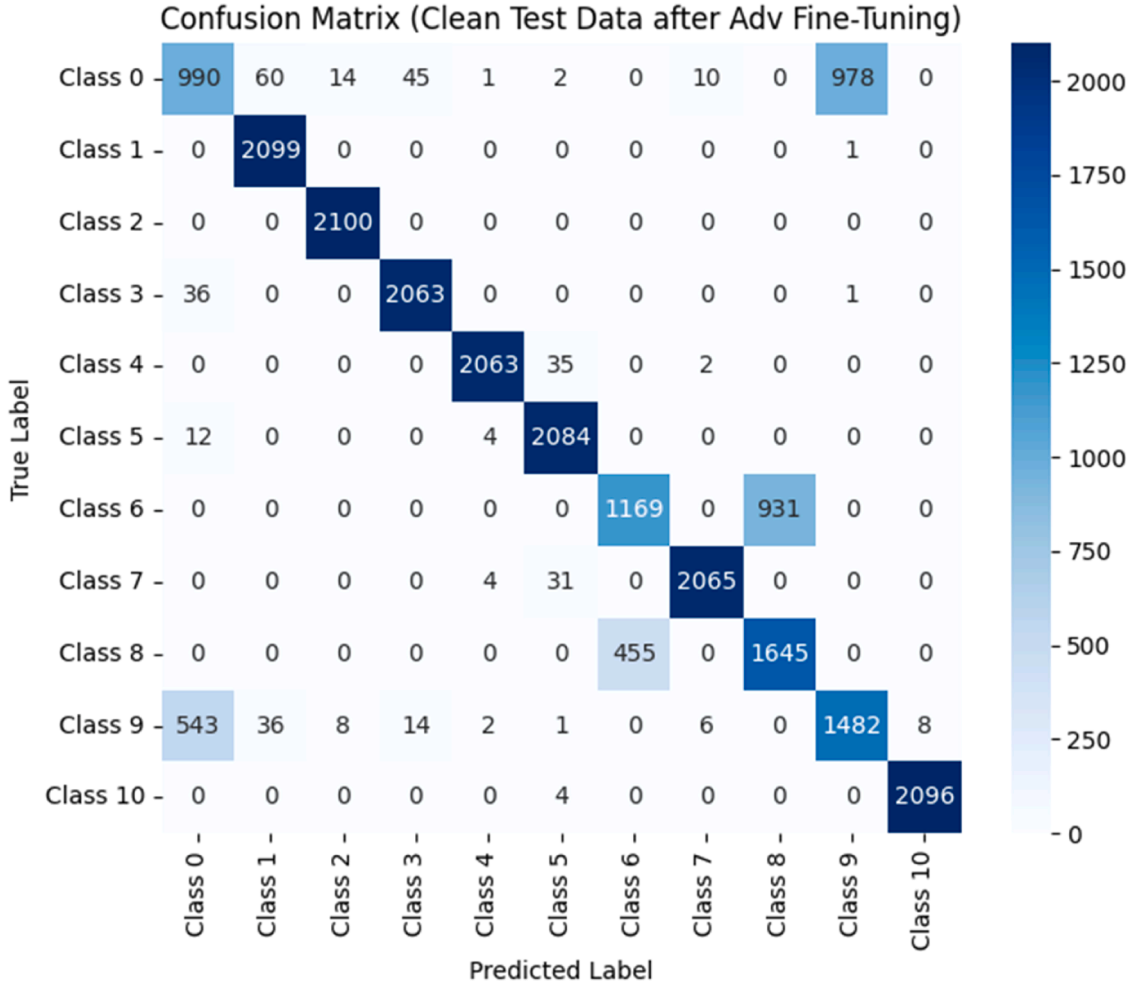


Fig. 11. Confusion matrix for proposed model.

4.2.1. Generalization across Noise mismatch

Fig. 7 illustrates the reconstruction error (MSE) and classification accuracy when model trained at fixed σ_{train} are evaluated across all σ_{eval} .

When the Joint DAE classifier is trained at a fixed noise scale σ_{train} and evaluated across varying evaluation noise levels σ_{eval} , we can analyze the model's ability to generalize under noise mismatch. Table 4 provides accuracies and reconstruction error of the Joint DAE classifier from $\sigma \in \{0-0.4\}$ under mismatched training and evaluation noise but model is trained at moderate noise $\sigma \in \{0.2-0.7\}$. Training noise levels are selected to signify moderate-to-strong privacy configurations during model development, whereas evaluation noise levels illustrate realistic deployment conditions where the effective noise level could decrease due to adaptive privacy policies, device limitations, or inference-time adjustments. This mismatch analysis evaluates the robustness and generalizability of the learned representations in the presence of unobserved noise conditions. The results show that the accuracy of the classification retains about the same for small to moderate amounts of evaluation noise ($\sigma_{eval} \leq 0.4$), independent of how much noise is present in the training set. This indicates that the DAE branch acquires a denoising capability that is applicable across various noise distributions. Reconstruction error, on the other hand, rises steadily as σ_{eval} increases, which means that stronger perturbations gradually change the learned feature feature representaions. Importantly, models trained at moderate noise levels $\sigma_{train} \in (0.2, 0.4)$ demonstrate the most balanced behavior: they preserve higher accuracy across mismatched settings while still achieving noticeable increases in reconstruction error, thus providing stronger privacy. These findings highlight the practical utility of training

with moderate DP noise, as it deploy the model to perform reliably even under deployment conditions where noise levels may differ from those seen during training.

4.2.2. Matched training & evaluation noise

Fig. 8 reports the results when training and evaluation are performed at the same noise scale. A second set of experiments examines the matched case, in which the noise from the training and evaluation processes is identical ($\sigma_{train} = \sigma_{eval}$). Table 5 provides accuracy and reconstruction error when training and evaluation use the same σ . The model is completely adapted to the distribution of noise encountered during inference in this scenario. Consequently, the accuracy is consistently superior to that of the mismatched case, particularly at low and moderate noise scales.

The controlled privacy-utility trade-off attained when the Joint DAE is optimized for the same perturbation level as evaluation is depicted by the gradual increase in reconstruction error with σ . Although privacy benefits are minimal, accuracy is nearly identical to the no-noise case at small noise scales. The classifier maintains competitive accuracy at moderate scales ($\sigma \approx 0.2 - 0.4$), while the reconstruction error increases sufficiently to provide meaningful distortion of activations, representing the optimal balance point. Nevertheless, the decoder's capacity to recover valuable features is diminished as σ increases beyond 0.5, as the reconstruction error increases sharply and the accuracy decreases more significantly. In the DP-only baseline, the reconstruction error rises from 5.53% to 11.0%, which is an absolute increase of about 5.5 percentage points. The reconstruction error changes from 1.97 to 3.33 in the DP +

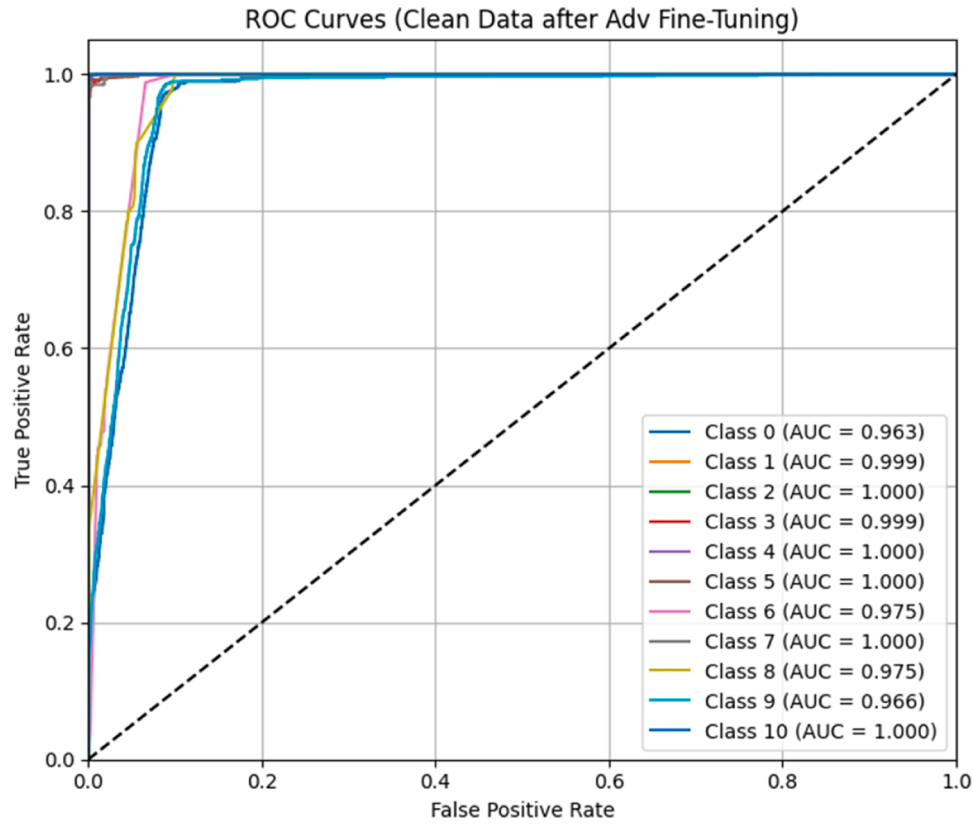


Fig. 12. ROC-AUC curves for proposed model.

Table 8

Comparison of proposed framework with DP-only and AT-Only models.

Variant	Clean Accuracy	FGSM ($\epsilon=0.05$)	PGD ($\epsilon=0.05$)	FGSM ($\epsilon=0.10$)	PGD ($\epsilon=0.10$)	Reconstruction MSE (%)	Key Benefit
DP-only(Joint DAE)	87.29	61.43	53.81	43.72	48.72	24.37	Privacy enhancement
AT-only(Joint DAE)	86.51	75.56	74.23	66.39	65.60	20.28	Robustness improvement
Proposed PrivRobust-SL (DP+AT)	85.2	74.22	73.10	65.34	64.75	34.12	Balanced Privacy-Utility-Robustness tradeoff

DAE setup, which is an absolute change of 1.36 and a relative increase of about 69%. The percentage change is greater in the DP + DAE configuration, but the absolute distortion remains within a lower-error range. This shows that semantic denoising can effectively stabilize noisy representations and preserves accuracy.

This analysis reiterates that the selection of σ remains crucial in determining the equilibrium between model utility and privacy guarantees, despite the fact that noise-aware training stabilizes performance under DP noise.

4.3. Vulnerability under Adversarial Attacks

After evaluating the differential-privacy behavior of the Joint DAE classifier, In this section we analysed its robustness to adversarial perturbations. The model was subjected to Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) attacks during this phase, without the use of any adversarial training. The model's vulnerability to crafted attacks was demonstrated, despite differential-privacy protection, as the perturbation strength (ϵ) was adjusted to simulate varying adversarial intensities. The adversarial examples were generated from the clean test set using FGSM and PGD with perturbation budgets $\epsilon \in \{0.01, 0.02, 0.05, 0.10\}$. Model parameters were kept fixed so that only the effect of adversarial perturbations on inference accuracy could

be measured. Fig. 9 shows the variation of accuracy for FGSM and PGD attacks across adversarial budget (ϵ).

The DP-trained Joint DAE classifier maintains a clear accuracy of 87.30, as evidenced by the results in Table 6, but it experiences a significant decline in accuracy as the intensity of the attack increases.

At $\epsilon = 0.05$, the accuracy of the FGSM decreases to 61.43, and at $\epsilon = 0.1$, it declines to 43.72. PGD demonstrates a comparable pattern, which indicates to its enhanced destructive capacity. These results illustrate that differential privacy does not inherently provide adversarial robustness, despite the fact that it provides feature-level protection. The motivation for implementing adversarial training in the subsequent phase to enhance robustness without compromising the protection of privacy is established by the significant decrease in both attack types.

4.3.1. Adversarial training for robustness

In this phase, the Joint DAE-classifier is fine-tuned using adversarial training (AT) to improve its resilience to gradient-based perturbations. Unlike the previous stage, where the model weights were fixed during attack evaluation, adversarial samples generated by FGSM and PGD are now included in the training loop. Each mini-batch contains a mixture of clean and adversarial inputs, enabling the model to learn perturbation and clean representations within its feature space. The adversarial training procedure follows a standard min-max formulation, where the

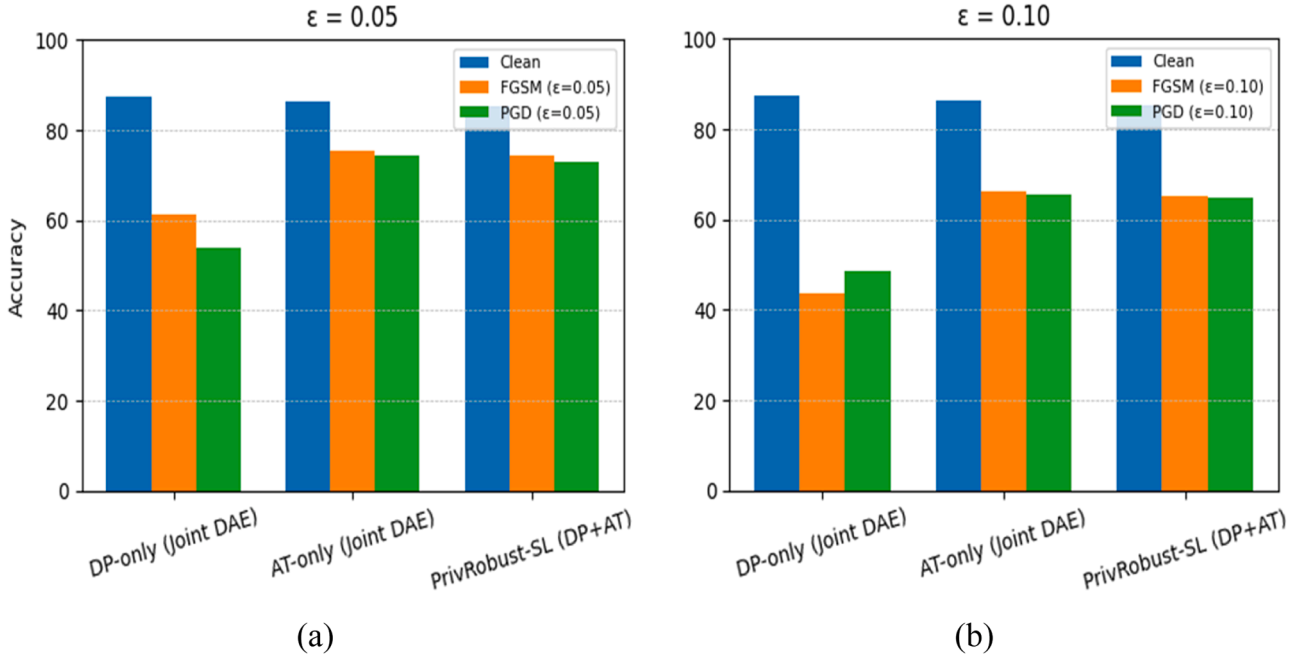


Fig. 13. Utility and robustness comparison (FGSM vs PGD).

model minimizes classification loss while maximizing the error caused by adversarial perturbations. The same hyper parameters and dataset splits as in the previous phase were maintained to ensure fair comparison. Table 7 presents accuracy due to FGSM and PGD attacks after adversarial training

Adversarial training yields a marked improvement in robustness. Clean accuracy is maintained at 87.30, while adversarial accuracy increases dramatically by over 10–20 percentage points at $\epsilon = 0.05$ for both FGSM and PGD. The model trained with AT learns smoother decision boundaries in the latent space, mitigating sharp gradient changes exploited by attacks. This demonstrates that integrating adversarial training with the Joint DAE architecture improves robustness to evasion attacks without introducing a significant degradation in clean data accuracy. Fig. 10 shows accuracy curves under FGSM and PGD perturbations before and after adversarial training. The adversarial training enhanced model maintains high accuracy even for larger ϵ , demonstrating its improved robustness.

Differential privacy lowers the risk of information loss, but it doesn't make systems fundamentally more resistant to attacks. The injected DP noise causes issues with intermediate representations and can make it harder to separate features, which makes the local gradients more sensitive. As a result, gradient-based attacks like PGD may work better in DP-only contexts by explaining degradation in adversarial robustness.

4.4. Proposed PrivRobust-SL (DP+AT with Joint DAE)

Building upon the preceding stages, the final proposed framework—PrivRobust-SL—integrates both Differential Privacy (DP) and Adversarial Training (AT) into the Split Learning architecture. Gaussian differential-privacy noise with a noise scale of $\sigma \in [0.0, 0.7]$ is introduced into the smashed activations on the client side before sending them to the server.

This lowers the possibility of inference-based privacy leakage by ensuring that critical input features are not exposed via activations, even in the event that an attacker has access to intermediate representations. The added noise works as a random stabilizer, pushing the model to learn representations that are more general and protect privacy. On the server side, the received noisy activations undergo joint optimization through the DAE and classifier branches. The DAE component

successfully regularizes the underlying feature space by attempting to recreate the original, clean activations from their noisy equivalents. In spite of privacy disruptions, this reconstruction effort aids the model in preserving representational accuracy. Concurrently, a mix of clean and adversarial samples produced using FGSM and PGD perturbations are used to train the classifier branch. These adversarial samples, which are created dynamically during each training cycle, are meant to resemble gradient-based attacks that attempt to mislead the classifier. This end-to-end joint optimization enables the model to simultaneously learn three key properties: (i) Noise tolerance through exposure to Gaussian perturbations (DP) (ii) Feature consistency via the DAE's reconstruction task, which anchors learning to stable latent representations (iii) Adversarial robustness by continuously adapting to FGSM/PGD-based gradient perturbations during training. During evaluation, the trained model is tested using both clean and adversarial inputs across different attack strengths (ϵ values). The adversarial samples are generated independently to ensure unbiased robustness testing.

The Confusion matrix of proposed adversarially trained split learning model incorporating differential privacy has demonstrated in Fig. 11. A notable disparity observed between overall accuracy and ROC-AUC curves show in Fig. 12. Adversarial fine-tuning and privacy-preserving techniques often smooth decision boundaries and add controlled noise, which might lower argmax-based accuracy on clean test data. However, the ROC-AUC curves remain high because they evaluate the threshold-independent quality of classification instead of fixed decision outcomes. Thereby, the high ROC-AUC values show that the model still has adequate probabilistic class separation and discriminative capacity, even though substantial misclassifications are present at the boundary level.

The combined DP + AT training allows the model to sustain high clean accuracy while retaining stability against both privacy attacks (reconstruction or inference) and adversarial evasion attempts. In the comparative summary Table 8, results are reported for $\epsilon = 0.05$ & 0.10 because these perturbation levels represents a balanced and standardized evaluation point widely adopted in adversarial-robustness literature. Smaller ϵ values (≤ 0.02) produce negligible distortions and are often visually imperceptible, leading to only marginal accuracy drops. Larger ϵ values (≥ 0.10) correspond to unrealistically strong perturbations that heavily degrade all models and obscure comparative insights.

Fig. 13 shows comparison of clean and adversarial accuracies (FGSM and PGD for $\epsilon = 0.05$ & $\epsilon = 0.10$) across DP-only, AT-only, and proposed PrivRobust-SL models. The proposed framework achieves improved robustness under both attacks with only a marginal reduction in clean accuracy, demonstrating its balanced privacy–utility–robustness trade-off. The reported findings can be interpreted as empirical observations, as the performance of the proposed framework may vary due to the dynamic nature of IoT traffic, its adaptability to different attack scenarios, and resource constraints.

5. Conclusion

This work presents PrivRobust-SL, a robust split learning framework designed to ensure privacy protection and adversarial robustness for IoT intrusion detection. The framework integrates client-side differential privacy with server-side adversarial training by employing joint denoising auto-encoder classifier architecture to enhance feature reconstruction and generalization. PrivRobust-SL effectively inhibits information leakage from intermediate representations while enhancing the system's resilience to adversarial perturbations. The combined optimization method enables the model to maintain robust detection performance despite the imposition of privacy and attack constraints. This demonstrates an effective balance among privacy, utility, and robustness. The experimental assessment shown that the incorporation of differential privacy-based noise and adversarial training not only safeguards data privacy but also enhances the network's stability and adaptability to various kinds of attack. More importantly, it improved adversarial accuracy by more than 15% at higher degrees of perturbation ($\epsilon = 0.10$) and maintained its performance well against both FGSM and PGD attacks. The proposed PrivRobust-SL framework emphasizes robustness and Privacy analysis instead of Communication overhead introduced by Adversarial training and model optimization techniques. The Exhaustive evaluation of proposed framework is limited due to variation of resource profiling across diverse attack scenarios and deployment platforms when used in constrained IoT applications.

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CRediT authorship contribution statement

V Santhosh Kumar: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Dhiraj Sunehra:** Writing – review & editing, Supervision, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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